

Experience:

Research intern | Agency for Science, Technology & Research (A-STAR) | Singapore 2025

- Developing a database linking data generated across L-PBF lifecycle at scale, using a postgresql backend and a django web frontend.
- Building prediction pipelines to extract insights from the database using Pytorch/sklearn libraries to analyze powder reusability for critical metal additive manufacturing applications.

Doctoral Candidate | University of Waterloo (MSAM Lab) | Canada 2021 - Present

- Using data-centric ML to enhance the generalizability of models developed for L-PBF additive manufacturing, specifically in the prediction of surface roughness of parts
- Building the largest reported dataset in literature of over 2000 surface roughness specimens
- Generating synthetic surfaces using ML predictions over L-PBF database using physics-informed ML and CAD handling libraries like numpy-stl and vtk.

Mechanical Test Engineer | Midmark Corporation, USA 2019 - 2020

- Led development of mechanical testing systems (~100) to validate crucial safety and functional needs of Class I/II medical devices.
- Analyzed and reported diverse testing data from sensors, failure tests, etc. using R+Shiny.

Manufacturing Engineer | Midmark Corporation, USA 2017-2019

- Conducted technical research and corporate approval for manufacturing equipment worth \$1.3M.
- Built a database of 5000 tool wear and breakages using VBA.
- Created a dashboard predicting tool breakage, saving ~\$50/breakage (avoiding ~80 breakages/month)

Education:

- PhD (Mechatronics Engineering), University of Waterloo, Canada | 93/100 **2021 - Present**
- MS (Industrial Engineering), TAMU, College Station, USA | 3.4/4.0 **2014-2016**
- BS (Production Engineering), GTU, India | 8.7/10.0 **2010-2014**

Honors and Awards:

- **MSFSS Award for research abroad** **2025**
- **Waterloo International Experience Award** **2025**
- **NSERC Canada Graduate Scholarship (CGS-D)** **2022-2025**
- **AMNOW data challenge by AmericaMakes- first place** **2022-2023**
- **President's Grad Scholarship Award** **2022-2025**
- **Faculty of Engineering Match Award** **2022-2025**
- **Ontario Graduate Scholarship (offered, declined)** **2022-2023**
- **Jim Rogers Scholarship by SME, Houston** **2016-2016**
 - Scholarship awarded for active contribution by students in the field of manufacturing.

Software Skills:

- **CAD/CAM:** working knowledge of Creo, Creo NC, SolidWorks, nTopology
- **Data sources:** RDBMS (postgres), API access, sensor-data streams
- **Data handling:** great expectations and pytest
- **ML modelling:** model development largely in Pytorch/sklearn
- **ML experimentation:** hydra, wandb, prefect, makefiles
- **General:** version control with Git/dvc, vtk for 3D object handling, Django for web, etc.

Publications and Presentations:

Journal Articles

1. **J. Patel**, M. Vlasea, S. Patel, “Towards generalizable machine learning prediction of downskin surface roughness in laser powder bed fusion”, *AIME 2025*. (Accepted, In-press)
2. **J. Patel**, M. Vlasea “LitDBr: Experimental Data on L-PBF surface roughness in literature”.
3. **J. Patel**, M. Vlasea “A data-driven review of experimental research in downskin surface roughness for L-PBF”. (*under preparation*)
4. **Patel J.B.**, Feng Z., Villanueva P.P, and Hung W.N.P., “Quality Enhancement with Ultrasonic Wave and Pulsed Current in Electrochemical Machining” *Procedia Manufacturing, Vol 10, 2017*, pp. 662- 673. Presented at NAMRC 2017.
5. NSLB Izwan, Z. Feng, **JB Patel**, WN Hung, “Prediction of Material Removal Rate in Die-Sinking Electrical Discharge Machining”, *Procedia Manufacturing, Vol 5, 2016*, pp. 658-668.

Conferences

1. **J. Patel**, M. Vlasea, “Data-Centric Machine Learning for Surface Roughness in Laser Powder Bed Fusion” in *RAPID+TCT- 2025*, Detroit, MI, USA, 2025
2. **J. Patel**, M. Vlasea, S. Patel, “Correlative Analytics of Downskin Surface Roughness Outcomes Using Systematic Data Science Workflows,” in *Holistic Innovation in Additive Manufacturing (HI-AM) Conference*, Halifax, NS, Canada. June 26-27,2023.
3. **J. Patel**, M. Vlasea, “Challenges in Physics-driven Machine Learning in Laser Powder Bed Fusion,” in *Holistic Innovation in Additive Manufacturing (HI-AM) Conference*, Montreal, QC, Canada. June 26-27,2022.
4. S. Patel, **J. Patel**, M. Yang, M. Vlasea, J. Barnes, C. Aldridge, W. King, J.Coyne., “Laser additive manufacturing of high reflectivity metallic materials using pore-free non-equiaxed powders,” in *RAPID+TCT - 2022*, Detroit, MI, USA, 2022.

Other Technical Papers

1. **J.Patel**, S. Patel, M. Vlasea, W. King, J. Barnes, C. Aldridge, “Toward improved process stability with low-cost non-spherical powder: a sensor driven case study”, for *Aubin AM Case Study Competition*, February 2023.
2. **J. Patel**, S. Patel, G. van Houtom, Z. Azimifar, M. Vlasea, *AMNOW Metals Challenge - Digital Thread Analysis for Build Improvements – Phase 2*, America Makes and NCDMM, 2022. **(Winner)**
3. **J. Patel**, S. Patel, G. van Houtom, Z. Azimifar, M. Vlasea, *AMNOW Metals Challenge - Digital Thread Analysis for Build Improvements – Phase 1*, America Makes and NCDMM, 2022. **(Winner)**